TITANIC DATASET

Exploratory Data Analysis

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GitHub Link	https://github.com/gauravyeskar/Titanic_EDA_Project

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1. Executive Summary

This report documents the end-to-end Data Analytic project containing data cleaning and preprocessing applied to the Titanic dataset as part of a professional Data analytics project. The raw dataset contained missing values in several key columns in numerical features that could negatively impact downstream modelling performance.

The preprocessing covered three major stages:

1. The null value analysis and imputation using a 5% threshold strategy.
2. Final Validation of the cleaned dataset.
3. Every decision was supported with visual comparisons (Kde, Boxplot, Bar chart) and statistical evidence (Standard Deviation Comparisons).

Key Outcomes:

- All missing values successfully imputed — zero nulls remain in the final dataset.
- Appropriate imputation strategy selected per column type and null percentage.
- Dataset Integrity preserved by not deleting a single row
- Appropriate feature engineering.

Metric	Before Cleaning	After Cleaning
Total Rows	1309	1309
Total Columns	12	18
Total Null Values	1698	0
Columns with Nulls	5	0

- Dataset is now clean, consistent, and ready for feature engineering and modelling.

Note: The total rows are considered by combining both the dataset.

2. Dataset Summary

The Titanic dataset is one of the most widely used datasets in data science and machine learning. It contains demographic and travel information about 1309 passengers aboard the RMS Titanic, along with a binary target variable indicating survival. Some rows are having missing survival rate. The survival column will show the distribution and will ask you to fill the survival column so that their distribution not change.

Column Description

	Type	Description	Nulls?
PassengerId	Numerical	Unique passenger identifier	No
Survived	Binary (0/1)	Survival flag — 0=No, 1=Yes	Yes
Pclass	Categorical	Ticket class — 1st, 2nd, 3rd	No
Name	Text	Full name of passenger	No
Sex	Categorical	Gender — male / female	No
Age	Numerical	Age in years	Yes
SibSp	Numerical	Number of siblings/spouses aboard	No
Parch	Numerical	Number of parents/children aboard	No
Ticket	Text	Ticket number	No
Fare	Numerical	Passenger fare paid	Yes
Cabin	Text	Cabin number	Yes
Embarked	Categorical	Port of embarkation — C, Q, S	Yes
Age Group	Categorical	Age Groups	Yes
Family Size	Numerical	Total Family Members	No
Surname	Text	Last Name	No
Title Grouped	Text	Mr. Miss. Mrs. Dr	No
Ticket Type	Text	Types of Tickets	No

Descriptive Statistics – Numerical Columns

The table below shows key statistical measures for all numerical columns in the raw dataset.

Column	Count	Mean	Std Dev	Min	25%	50%	75%	Max
Age	1046	29.00	14.41	0.17	21.00	28.00	39.00	80.00
Fare	1308	33.29	51.75	0.00	7.89	14.45	31.27	512.32

Column	Count	Mean	Std Dev	Min	25%	50%	75%	Max
SibSp	1309	0.49	1.04	0.00	0.00	0.00	1.00	8.00
Parch	1309	0.38	0.86	0.00	0.00	0.00	0.00	9.00

3. Null Value Summary

Before handling missing data, a thorough null analysis was performed to understand the scale and distribution of missing values. Columns were classified into two groups based on a 5% threshold a standard industry practice that balances data quality with distribution preservation.

Null Value Summary

Column	Null Count	Null %	Group	Strategy
Age	263	20.09 %	More than 5%	Interactive — user selected
Cabin	1014	77.46%	More than 5%	Interactive — user selected
Embarked	2	0.15 %	Less than 5%	Auto-filled with Mode
Survived	418	31.93%	More than 5%	Interactive — user selected

4. Null Handling — Columns with Less Than 5% Nulls

Columns with 5% or fewer missing values were automatically filled without user intervention. Since the missing proportion is small, the impact on overall distribution is minimal. The following strategies were applied:

• Continuous numerical columns — filled with the Mean (average value).

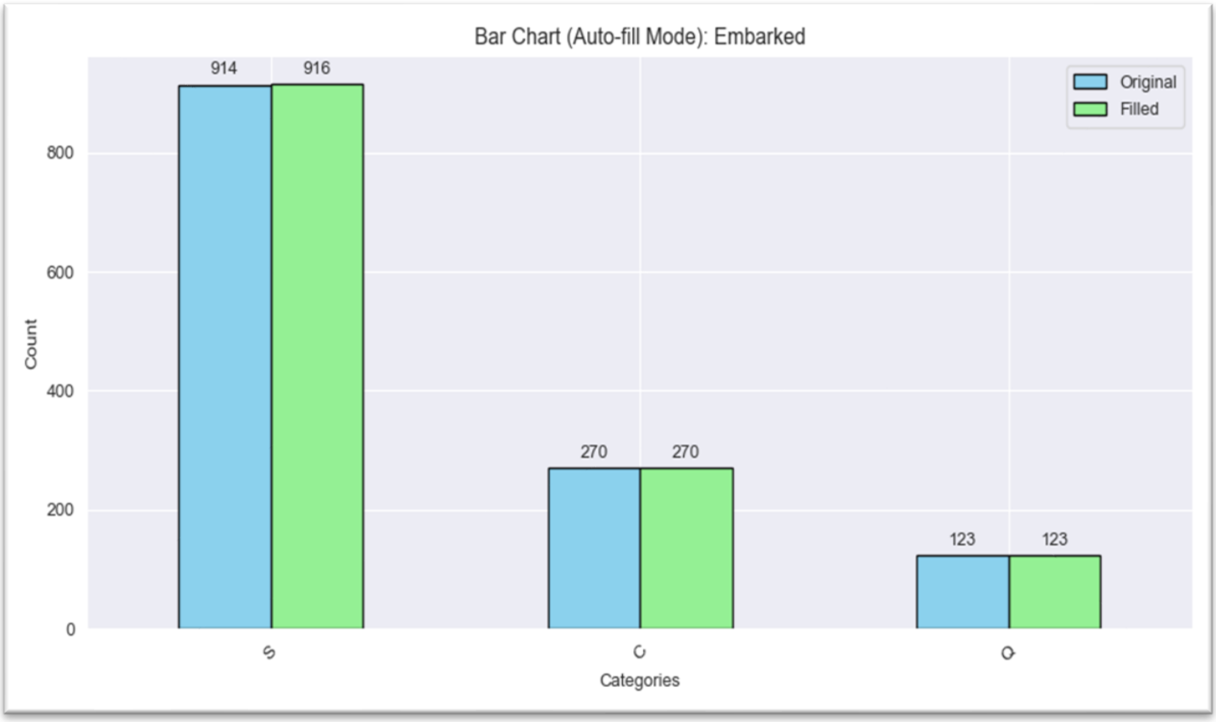
• Binary / low-cardinality numerical columns — filled with the Mode (most common value).

• Text / categorical columns — filled with the Mode (most frequent category).

Column: Embarked (Distribution Comparison – embarked (Before / After) filling**)**

Property	Value
Null Count	2
Null %	0.22%
Column Type	Categorical

Property	Value
Strategy Used	Mode Fill
Mode Value	S (Southampton)



Observation: Filling 2 rows with Mode = 'S' has a negligible effect on the overall category distribution since it represents only 0.22% of the data.

4. Null Handling — Columns with More Than 5% Nulls

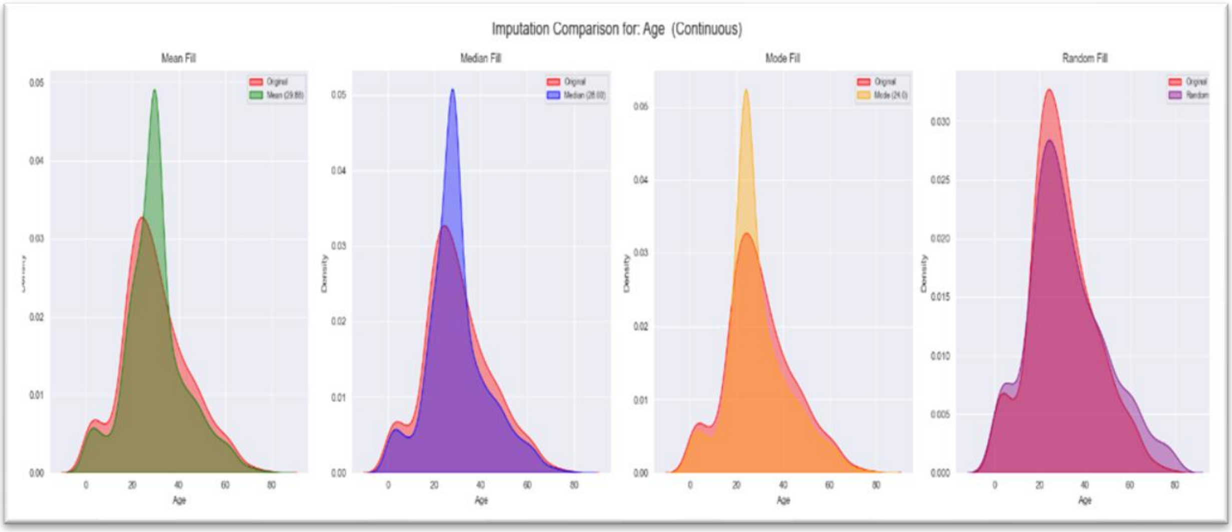
Columns with more than 5% missing values require careful treatment to avoid distorting the data distribution. For each such column, four fill strategies were compared visually and statistically before a final choice was made. When you run the program, you will get to know about all the methods we have implemented in it.

Strategy	Method	Best For	Risk
Mean	Fill with average value	Symmetric distributions	Reduces variance
Median	Fill with middle value	Skewed distributions	Slight variance loss
Mode	Fill with most common value	Binary / categorical	Inflates one class

Strategy	Method	Best For	Risk
Random	Fill with random in-range values	Preserving spread	Adds randomness

Column: Age (20.09% Nulls)

Age is a continuous numerical column with a moderately right-skewed distribution. Four strategies were compared using KDE plots and standard deviation analysis.



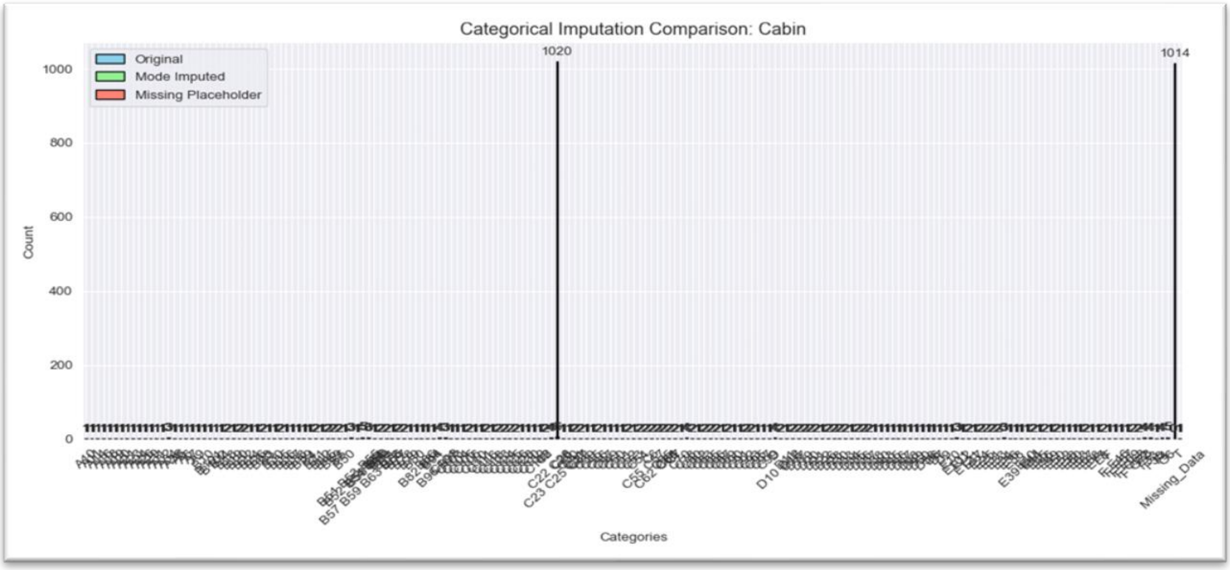
Standard Deviation Comparison — Age

Strategy	Std Dev	Difference from Original	Verdict
Original (with nulls)	14.41	—	Baseline
Mean Fill	12.88	1.12	Slightly reduces spread
Median Fill	12.09	1.91	Slight difference to mean
Mode Fill	13.09	0.91	Best — preserves variance
Random Fill	17.02	-3.02	Inflates on age group

Decision: [Write your chosen strategy here — e.g.: Mode fill was chosen as it best preserved the distribution shape for the Age column.']

Column: Cabin (77.46% Nulls)

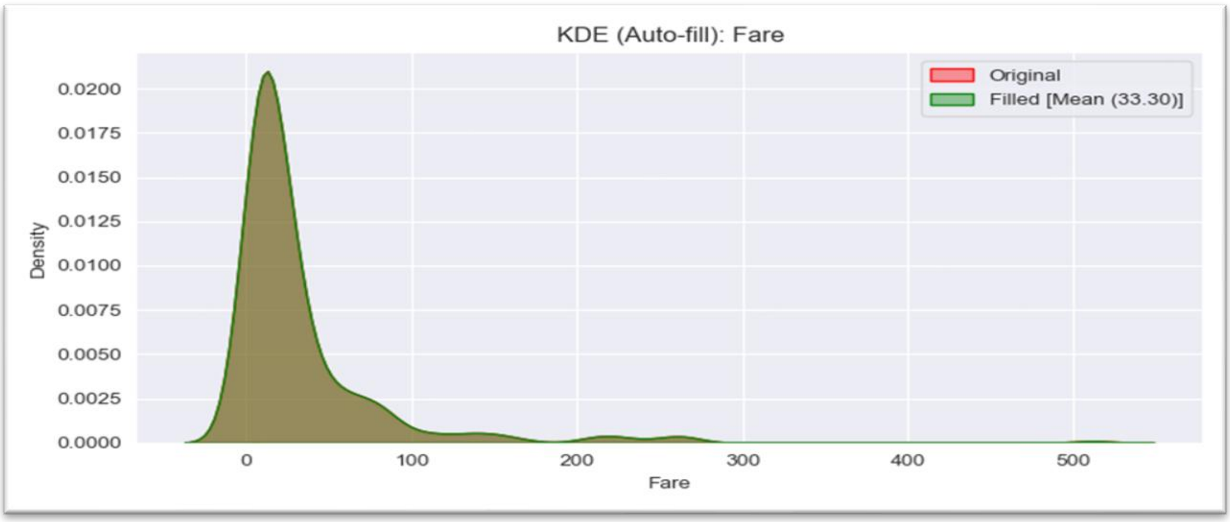
The Cabin column has an extremely high null rate of 77.46%. This is a text column where the actual cabin number is missing for most passengers. Two strategies were compared.



Decision: Due to the very high null percentage (77.46%), filling with Mode would artificially inflate one cabin value. 'Missing Data' placeholder was used to acknowledge the absence of information without introducing bias.

Column: Fare (0.07% Nulls)

The Fare column has less than 5% missing values so we auto filled with the mean value.



5. Final Dataset Status

After completing all cleaning steps, the dataset was validated to confirm zero null values remain and all column distributions are consistent with the original data.

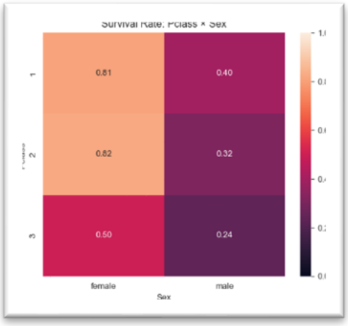
Null Check After Cleaning

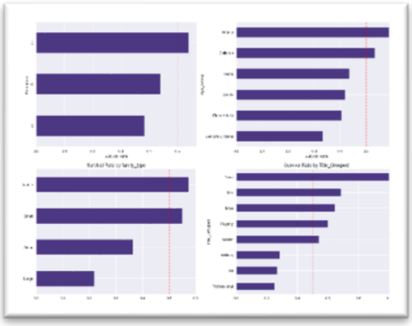
Column	Nulls Before	Nulls After	Status
Age	263	0	Clean
Cabin	1014	0	Clean
Embarked	2	0	Clean
Survived	418	0	Clean
Fare	1	0	Clean

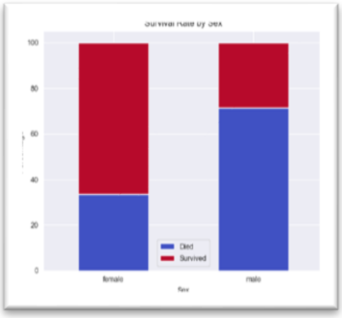
Final Dataset Shape

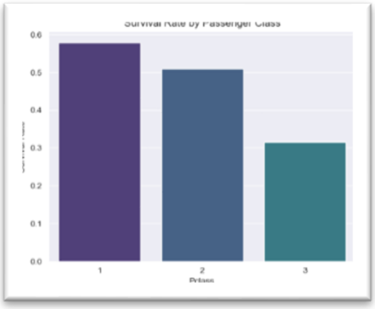
Property	Value
Total Rows	1309
Total Columns	18
Null Values Remaining	0
Rows Removed	0
Columns Removed	0

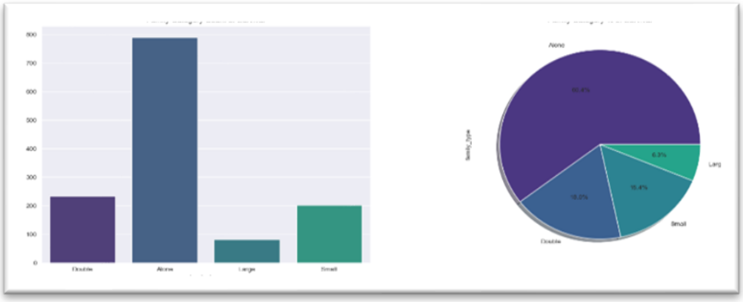
Final Outputs

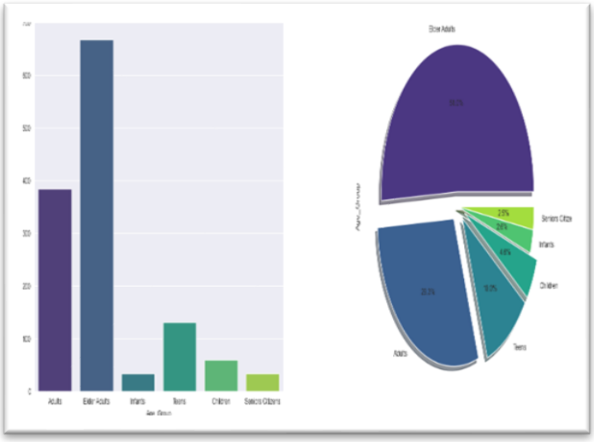


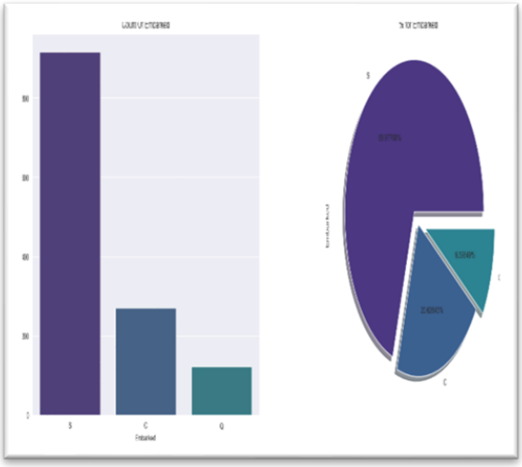


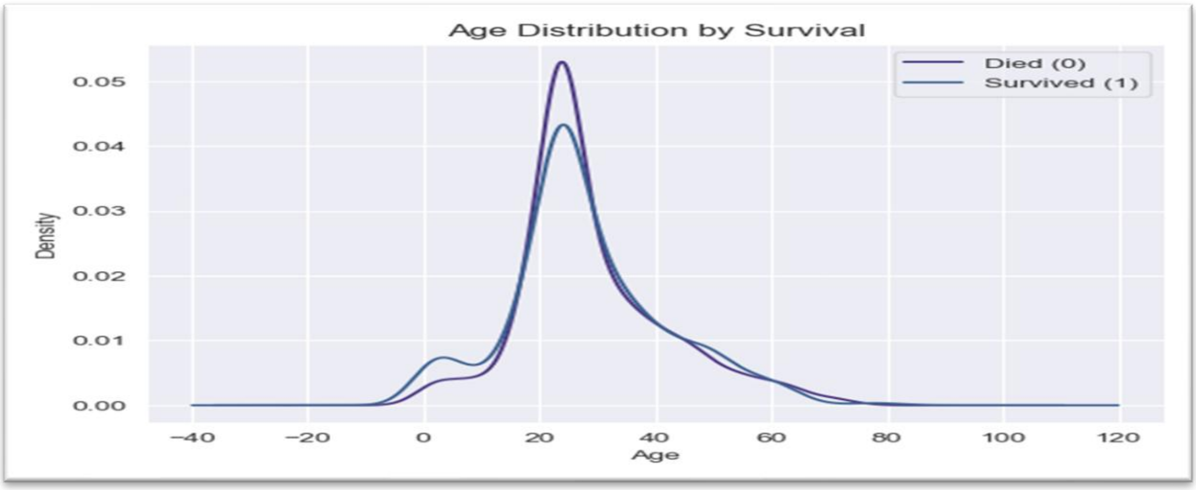


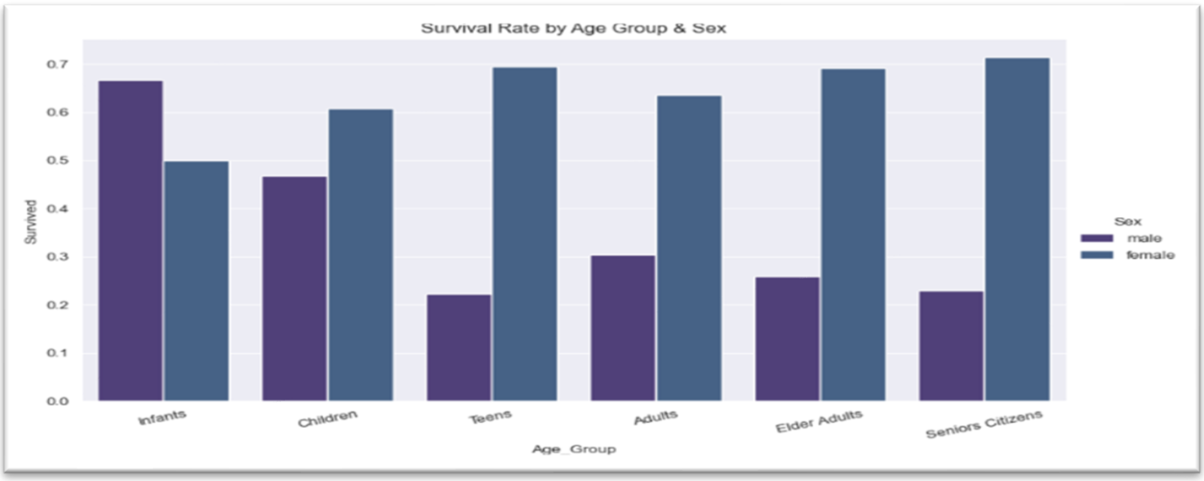

0.6
0.5
0.4
0.3
0.2
0.1
0.0
1
2
3



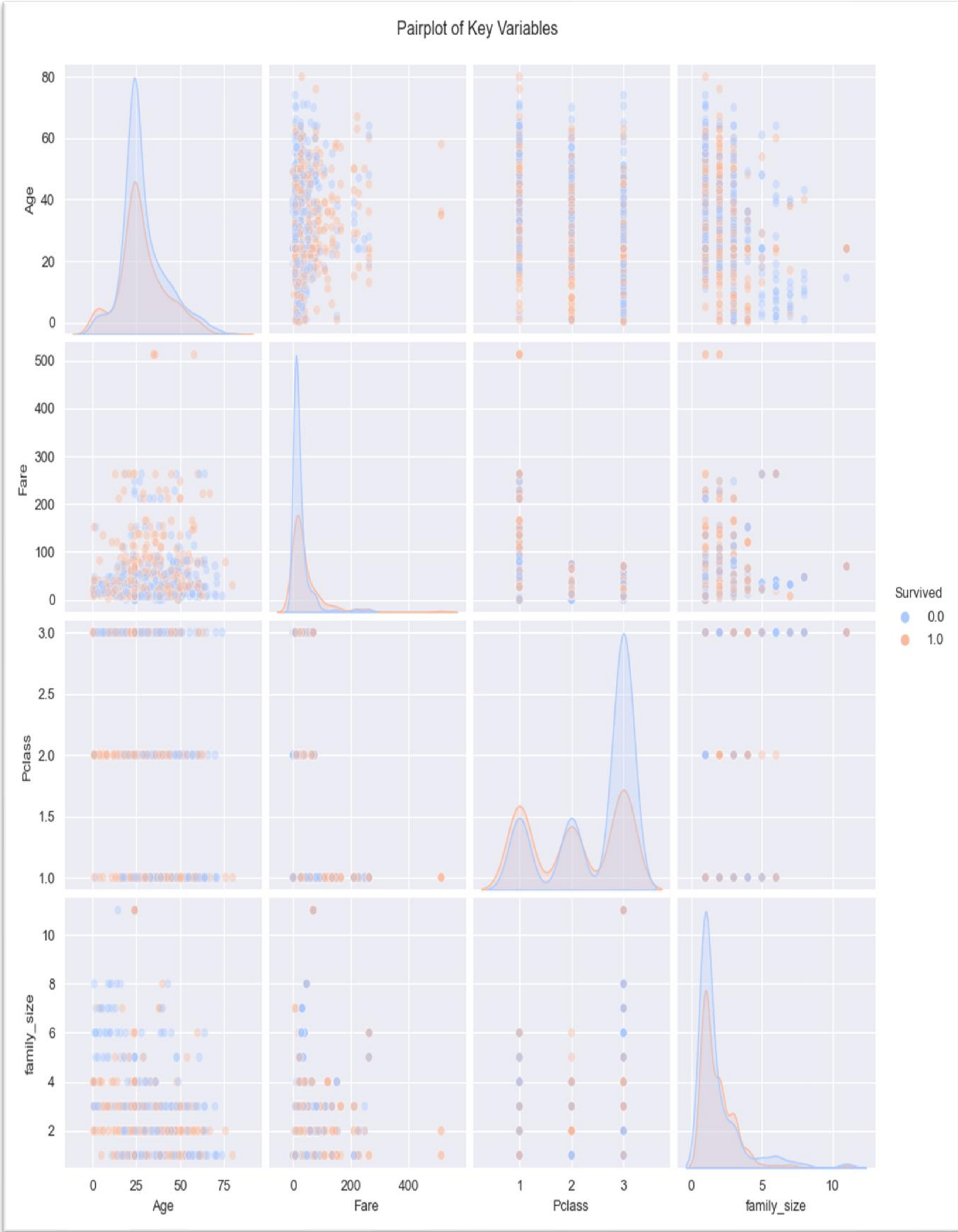





Age Distribution by Survival
Died (0)
Survived (1)
Density
0.05
0.04
0.03
0.02
0.01
0.00
−40
−20
0
20
40
60
80
100
120
Age


Survival Rate by Age Group & Sex
Survived
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Infants
Children
Teens
Adults
Elder Adults
Seniors Citizens
Age_Group
Sex
male
female

Complete Analysis at One Place



6. Conclusion & Recommendations

Conclusion

The Titanic dataset has been successfully cleaned and pre-processed. The pipeline followed industry-standard practices for handling missing values and end-to-end analysis, with every decision validated using visual and statistical evidence.

Based on my analysis –

- Males & Females of 1st class was saved the most followed by 2nd and 3rd.
- Southampton (S) Embarked was saved mostly.
- Infants were saved in that too Female infants were saved the most.
- The passengers travelling alone and with someone (2) were safe.
- The passengers between the age 15 to 50 survived the most.
- 1st Class Passengers survived the most.
- The passengers who has not paid a single dollar deceased the most.

Recommendations

- Based on the analysis:

  - Emergency evacuation procedures should prioritize vulnerable groups such as children and elderly passengers.

  - Equal access to lifeboats should be ensured across all passenger classes.

  - Passenger safety training should be standardized regardless of ticket class.

  - Capacity planning should account for family groups because passengers traveling with small families showed higher survival rates.